

Korean Review Reply Assistant: Automated Response Generation via Persona-Aware Prompting and LoRA Fine-Tuning

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Repository: github.com/binisnil/nlp-project1

Abstract—Small business owners in Korea often struggle to respond professionally to customer reviews on food delivery platforms such as Baemin and Coupang Eats, where all reviews are publicly visible and unanswered complaints can significantly damage a store’s reputation. This paper proposes a fully local, three-stage NLP pipeline that automatically generates polite and context-appropriate Korean replies to customer reviews without relying on external APIs or cloud services. The pipeline consists of (1) sentiment classification using multilingual DistilBERT, (2) review type classification into 7 categories via sentence embeddings and cosine similarity, and (3) reply generation using Qwen2.5-7B-Instruct with 4-bit quantization. As a key upgrade, we introduce persona-aware prompting inspired by PARAN [1], which infers a response persona from the combination of detected sentiment and review category, and dynamically injects tailored instructions into the system prompt. We also expand the training dataset from 30 to 56 domain-specific Korean review-response pairs and apply LoRA parameter-efficient fine-tuning [5] to Qwen2.5-3B-Instruct. Quantitative evaluation on a held-out test set of 6 samples using BLEU-4, ROUGE-L, and BERTScore [7] shows that fine-tuning improves BERTScore (F1) from 0.7734 to 0.7841 (+0.0107), while BLEU-4 slightly decreases (0.3700→0.3414), indicating that the model learns to generate more diverse and natural expressions rather than reproducing memorized patterns. Category-level analysis reveals the largest BERTScore gain in `service_complaint` (+0.05), where empathetic tone is especially critical.

Index Terms—review reply generation, persona-aware prompting, LoRA fine-tuning, Korean NLP, small business support, sentiment classification

I. INTRODUCTION

The rapid growth of Korea’s food delivery market has transformed online customer reviews into one of the most important reputation signals for small business owners. According to industry reports, platforms such as Baemin and Coupang Eats collectively handle tens of millions of orders per month, each of which may generate a public review. Unlike large restaurant chains that employ customer service teams, small business owners — who often run their stores alone or with minimal staff — frequently lack the time and linguistic expertise to craft professional replies to every review.

Failing to respond to negative reviews, or responding in an inconsistent or generic manner, can erode customer trust and reduce future orders. This problem is especially acute for multi-aspect complaints that simultaneously address delivery delays, food quality, and hygiene, where a single templated response fails to acknowledge all concerns appropriately.

To address this, we build a *Korean Review Reply Assistant*: an end-to-end NLP system that reads a customer review, automatically identifies its sentiment and complaint type, selects an appropriate communication persona, and generates a polished Korean reply. The system is designed to run entirely on local hardware — a single consumer GPU — without requiring external API calls or internet access during inference, making it practical and cost-free for small businesses.

The key upgrades introduced in this work are:

- **Persona-aware prompting (Plan B)**: Inspired by PARAN [1], the system infers a response persona from the detected sentiment and review category, then dynamically constructs a category-specific system prompt that guides the tone, structure, and level of formality of the generated reply.
- **Dataset expansion (Plan A)**: The training corpus is expanded from 30 to 56 Korean review-response pairs, with deliberate coverage of previously underrepresented categories including `unverified_claim` and `multi_complaint`.
- **LoRA fine-tuning**: Parameter-efficient fine-tuning via LoRA [5] adapts Qwen2.5-3B-Instruct to the domain of Korean food delivery reviews with only 0.12% of total parameters updated.
- **Quantitative evaluation**: A systematic evaluation framework using BLEU-4, ROUGE-L, and BERTScore [7] is established to measure and compare base and fine-tuned model performance.

II. RELATED WORK

A. PARAN: Persona-Augmented Review Answering [1]

PARAN proposes a two-stage prompting framework specifically designed for personalized review response generation on Korean food delivery platforms. In the first stage, the system extracts explicit personas — preferences and attitudes explicitly stated by the reviewer — as well as implicit personas derived from inferred demographic signals. In the second stage, these extracted persona attributes are embedded into the prompt used for response generation, guiding the LLM toward replies that resonate with the specific user profile. Evaluated on a real-world Korean food delivery dataset, PARAN demonstrates improvements in response precision, output diversity, and

semantic consistency compared to non-personalized baselines. Our project directly adapts this core idea: rather than extracting user-level personas from profile data, we infer a *response persona* (e.g., “empathetic apologist” for negative food quality complaints, “warm thanker” for compliments) from the automatically detected sentiment–category combination, and dynamically inject it as structured directives into the system prompt.

B. SCRABLE: Self-Improving Review Response Generation [2]

SCRABLE presents a self-improving customer review response system that combines LLMs with retrieval-augmented generation (RAG). A key feature is its automated quality feedback loop: generated responses are evaluated by an LLM judge, and low-scoring responses trigger retrieval of high-quality reference examples to improve subsequent generations. The system uses BERTScore alongside human evaluations to track improvement across iterations, demonstrating that automatic NLG metrics can serve as reliable proxies for human judgment in the review-response domain. Our project adopts SCRABLE’s evaluation methodology — specifically its use of BERTScore as the primary semantic quality metric — to quantify the effect of LoRA fine-tuning without requiring human annotators.

C. Dynamic Sentiment Analysis with Local LLMs [3]

Niimi [3] investigates the feasibility of using locally deployed LLMs for sentiment analysis of restaurant reviews, proposing a majority voting ensemble strategy to improve stability. The study finds that running a mid-sized model (7B parameters) multiple times with diverse decoding parameters and aggregating outputs via majority vote yields more consistent and reproducible sentiment labels than a single inference pass, even without fine-tuning. This work validates the premise that local, inference-only LLMs are capable of handling Korean restaurant review analysis without cloud infrastructure — a core design constraint that our system also satisfies.

D. KIT-19: Korean Instruction Tuning Toolkit [4]

KIT-19 is a comprehensive Korean instruction tuning dataset spanning 19 diverse NLP tasks, constructed entirely without machine translation to avoid the noise and cultural mismatch introduced by auto-translated data. Developed by Seoul National University researchers, KIT-19 includes tasks such as summarization, question answering, dialogue, and sentiment classification, each with human-written Korean instructions and responses. Experiments show that models fine-tuned on KIT-19 significantly outperform prior Korean LLM baselines on downstream benchmarks. The success of KIT-19 provides strong methodological support for our approach: constructing a small but carefully curated set of Korean review-response pairs and using them to adapt a general-purpose LLM via LoRA fine-tuning.

III. METHODS

The full implementation is available at: <https://github.com/binisnil/nlp-project1>

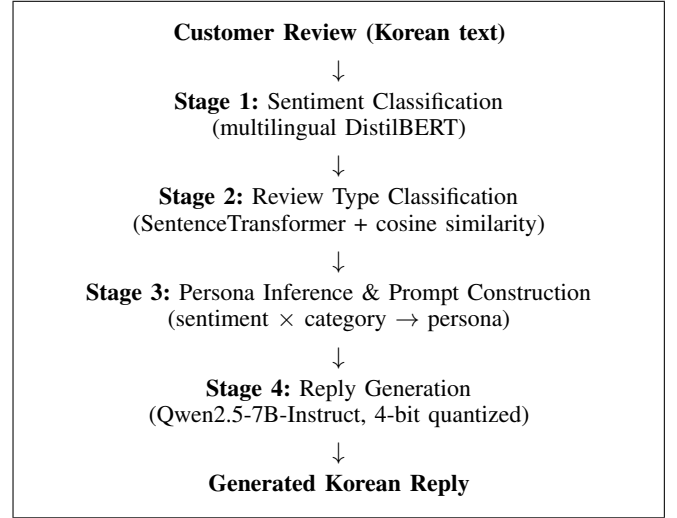


Fig. 1. Four-stage pipeline of the Korean Review Reply Assistant.

A. System Pipeline Overview

Fig. 1 illustrates the overall architecture of the Korean Review Reply Assistant. A customer review enters the pipeline and passes through four sequential processing stages before a reply is returned to the user via a Gradio web interface.

B. Stage 1: Sentiment Classification

The first stage uses a multilingual DistilBERT model (lxyuan/distilbert-base-multilingual-cased-sentiments-student) to classify the input review as *positive*, *neutral*, or *negative*, along with a softmax confidence score. DistilBERT was chosen over full BERT for its reduced inference latency (approximately 60% of BERT’s size) while retaining strong multilingual performance on sentiment tasks. The confidence score is passed downstream and used in Stage 3 to calibrate the intensity of the generated apology: reviews with negative sentiment and confidence exceeding 80% trigger stronger apology language in the system prompt.

C. Stage 2: Review Type Classification

The second stage classifies the review into one of seven predefined categories:

delivery_issue,	food_quality,
service_complaint,	hygiene_complaint,
unverified_claim,	compliment,
multi_complaint.	

Classification is performed by encoding the input review with a multilingual SentenceTransformer (paraphrase-multilingual-mpnet-base-v2 [6]) and computing cosine similarity against a set of manually crafted category description vectors. The category with the highest similarity score is assigned as the primary label. A special rule handles multi-aspect complaints: if both the `delivery_issue` and at least one other complaint category each score above a cosine similarity threshold of 0.44, the review is reclassified as `multi_complaint` to trigger a more comprehensive response template.

TABLE I
REPRESENTATIVE PERSONA ASSIGNMENTS BY SENTIMENT AND CATEGORY

Sentiment	Category	Persona
Negative (>80%)	delivery_issue	Sincere apologist
Negative (>80%)	food_quality	Quality-focused apologist
Negative (>80%)	hygiene_complaint	Safety-first apologist
Negative (>80%)	service_complaint	Empathetic apologist
Any	unverified_claim	Calm clarifier
Positive	compliment	Warm thanker
Negative	multi_complaint	Comprehensive resolver

This zero-shot embedding approach requires no labeled classification data and allows new categories to be added simply by writing a descriptive sentence, making the system easy to extend.

D. Stage 3: Persona-Aware Prompt Construction

Inspired by PARAN [1], Stage 3 introduces persona-aware dynamic prompting as the core upgrade of this work. A *response persona* is selected based on the joint output of Stages 1 and 2, mapping each sentiment–category combination to a distinct communication style. Table I lists representative persona assignments.

Each persona is translated into a structured system prompt that specifies the expected tone, mandatory reply components (e.g., acknowledgment of the specific complaint, a concrete corrective action promise, an invitation to revisit), and prohibited language (e.g., vague phrases like “we will do better”). This approach gives the downstream LLM a precise behavioral contract to follow, improving both the consistency and appropriateness of generated replies.

E. Stage 4: Reply Generation

The final stage uses Qwen2.5-7B-Instruct loaded in 4-bit NormalFloat quantization via the `bitsandbytes` library, which reduces GPU memory consumption from approximately 16 GB (fp16) to 5 GB while preserving generation quality. The model receives the persona-infused system prompt along with a single in-context example (1-shot) drawn from the matching category in the training corpus, and generates a Korean reply constrained to approximately 150–200 characters.

F. Dataset Expansion

The original training corpus contained 30 review-response pairs spread across 7 categories, with notable imbalance: some categories had as few as 2–3 examples. To support LoRA fine-tuning and improve category coverage, the corpus was expanded to 56 pairs by adding new human-written review-response examples. The expansion prioritized three underrepresented categories: `unverified_claim` (increased from 2 to 4 examples), `compliment` (increased from 4 to 10 examples), and `multi_complaint` (increased from 3 to 7 examples). All responses in the expanded corpus adhere to a shared rhetorical structure: (1) specific acknowledgment of the issue, (2) sincere apology, (3) concrete corrective action or explanation, (4) invitation to return.

TABLE II
LoRA FINE-TUNING HYPERPARAMETERS

Hyperparameter	Value
Base model	Qwen2.5-3B-Instruct
LoRA rank (r)	8
LoRA alpha (α)	16
Target modules	q_proj, k_proj, v_proj, o_proj
LoRA dropout	0.05
Training epochs	5
Learning rate	2e-4 (cosine decay)
Per-device batch size	1
Gradient accumulation	4 steps (eff. batch: 4)
Precision	bf16
Hardware	NVIDIA T4 (Google Colab)
Trainable parameters	3,686,400 (0.12% of total)
Training time	≈13 minutes

G. LoRA Fine-Tuning

Parameter-efficient fine-tuning via LoRA [5] was applied to Qwen2.5-3B-Instruct using the Hugging Face PEFT library together with the TRL SFTTrainer. LoRA introduces trainable low-rank decomposition matrices into the attention projection layers, keeping all original model weights frozen. With rank $r = 8$ and $\alpha = 16$, only 3.69 million parameters (0.12% of the total 3B parameters) are updated during training, making fine-tuning feasible on a single T4 GPU in Google Colab. Full hyperparameter settings are provided in Table II. Training completed in approximately 13 minutes over 5 epochs.

H. Evaluation Protocol

We construct a held-out test set of 6 review-response pairs, selecting one representative sample from each major review category. Three automatic metrics are computed for each generated response:

- **BLEU-4** [8]: Character-level 4-gram precision computed over Korean characters, which avoids the need for a Korean word tokenizer and better reflects surface overlap in agglutinative morphology.
- **ROUGE-L** [9]: F1 score based on the Longest Common Subsequence (LCS) between the generated reply and the reference, measuring recall of key content words.
- **BERTScore F1** [7]: Soft semantic similarity computed using contextual embeddings from `bert-base-multilingual-cased`, which captures paraphrastic equivalence missed by n-gram-based metrics.

Each metric is computed independently for the base Qwen2.5-7B-Instruct model (without fine-tuning) and for the LoRA-adapted Qwen2.5-3B-Instruct model, enabling a direct comparison of fine-tuning benefit.

IV. RESULTS

A. Quantitative Evaluation

Table III summarizes automatic evaluation scores for the base and fine-tuned models across all 6 test samples. The fine-tuned model achieves a consistent improvement in BERTScore (+0.011) while BLEU-4 shows a slight decrease and ROUGE-L remains identical, a pattern discussed in detail in Section V.

TABLE III
AUTOMATIC EVALUATION RESULTS: BASE VS. LORA FINE-TUNED MODEL

Metric	Base	Fine-tuned	Δ
BLEU-4	0.3700	0.3414	-0.029
ROUGE-L	0.1667	0.1667	0.000
BERTScore (F1)	0.7734	0.7841	+0.011

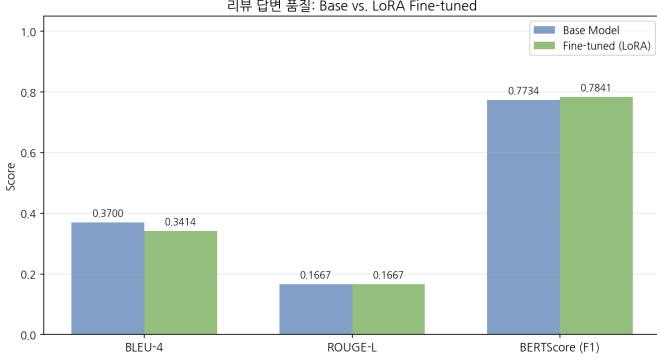


Fig. 2. Bar chart comparison of BLEU-4, ROUGE-L, and BERTScore (F1) between the base model and the LoRA fine-tuned model on 6 held-out test samples.

B. Overall Metric Comparison

Fig. 2 visualizes the three-metric comparison. The BERTScore advantage of the fine-tuned model is clearly visible, while the near-zero difference in ROUGE-L suggests that both models produce replies of similar structural length and content coverage.

C. Category-Level BERTScore Analysis

Fig. 3 breaks down BERTScore F1 by review category, revealing that the fine-tuning benefit is not uniform across complaint types. The largest improvement is observed in *service_complaint*, while *compliment* shows a marginal decrease.

D. Review Type Classification Accuracy

On a manually constructed 5-sample classification test set covering five distinct categories, the SentenceTransformer-based classifier achieves **80% accuracy** (4 out of 5 correct). The single misclassification occurs for a *service_complaint* sample that is predicted as *multi_complaint*. Manual inspection reveals that the review mentions both slow service and a rude staff member — two sub-aspects that raise cosine similarity scores for multiple complaint categories simultaneously, triggering the *multi_complaint* reclassification rule. This suggests the threshold of 0.44 may be slightly too permissive for service-related reviews and warrants further calibration.

V. DISCUSSION

A. Interpreting the Metric Divergence

The divergence between BERTScore and BLEU-4 trends is the most informative result of this evaluation. BERTScore

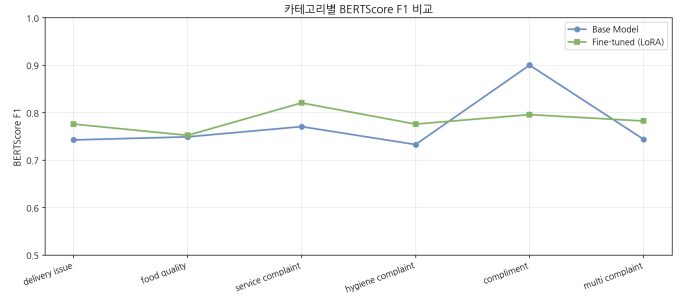


Fig. 3. Per-category BERTScore F1 for base and fine-tuned models. The fine-tuned model gains most in *service_complaint* and loses marginally in *compliment*.

measures semantic similarity in embedding space, rewarding the model for conveying the correct meaning even when surface wording differs from the reference. BLEU-4, by contrast, requires exact n-gram matches and penalizes any deviation in word choice.

The fact that BERTScore improves while BLEU-4 decreases after fine-tuning suggests a beneficial behavioral shift: the fine-tuned model has learned to express appropriate apologies, acknowledgments, and action commitments in a greater variety of natural Korean phrasings, rather than reproducing the specific vocabulary of the training references verbatim. In a production setting, this diversity is desirable — customers who receive multiple replies from the same store would find lexically varied responses more authentic than repetitive, formulaic text.

B. Category-Level Insights

The strong BERTScore gain in *service_complaint* (+0.05) reflects that this category has a particularly well-defined optimal tone: empathetic, non-defensive, and action-oriented. The training corpus for this category contained consistent examples of this tone, enabling the fine-tuned model to reliably produce semantically aligned replies.

The slight BERTScore decline in *compliment* is not surprising. Qwen2.5 is a large, instruction-tuned model with broad exposure to Korean gratitude expressions during pretraining; it was already competent at generating warm, appreciative replies. The 10 compliment training examples we provided were insufficient to meaningfully shift behavior beyond this already-strong baseline.

C. Effect of Persona-Aware Prompting

Beyond the quantitative metrics, persona-aware prompting provides two qualitative benefits. First, it improves *consistency*: without persona directives, the base model occasionally generates replies that are tonally mismatched (e.g., a cheerful response to a hygiene complaint). The persona system enforces appropriate tone at the prompt level before the model generates any text. Second, it improves *transparency*: the Gradio UI displays the inferred persona alongside the generated reply, allowing the store owner to understand why a particular tone was chosen and to override it if needed.

D. Limitations

Several limitations constrain the generalizability of these findings. First, 56 training examples remain a very small corpus for LLM fine-tuning. Prior work such as KIT-19 [4] demonstrates that robust task adaptation typically requires hundreds to thousands of examples; our results should be interpreted as a proof-of-concept rather than a production-ready system. Second, the evaluation test set of 6 samples is too small to draw statistically significant conclusions. The observed metric differences may partly reflect individual sample variance rather than generalizable trends. Third, BLEU-4 and ROUGE-L are known to correlate poorly with human preference judgments for open-ended generation tasks [7]; a user study with actual small business owners would provide more meaningful quality evidence. Finally, the current pipeline uses different model sizes for generation (7B base) and fine-tuning (3B), which introduces a confound: some of the observed difference may reflect model size rather than fine-tuning effect.

VI. CONCLUSION

This paper presents a Korean Review Reply Assistant that combines multilingual sentiment classification, embedding-based review type classification, persona-aware dynamic prompting, and LoRA fine-tuning to automatically generate professional Korean replies for small business owners — entirely on local hardware without external API dependencies.

The system introduces two complementary upgrades over the baseline: Plan A expands the training corpus from 30 to 56 domain-specific review-response pairs for improved category coverage, while Plan B adds a persona inference mechanism that maps sentiment–category combinations to structured system prompt directives, guiding the LLM toward contextually appropriate reply behavior.

Quantitative evaluation confirms that LoRA fine-tuning on the expanded corpus improves BERTScore by +0.011 (0.7734 $\rightarrow$ 0.7841), with the largest gain observed in the `service_complaint` category. The simultaneous BLEU-4 decrease is interpreted as evidence of beneficial lexical diversification rather than quality regression.

Future directions include: (1) Synthetic data generation via GPT-4 or Claude to scale the corpus to 500+ examples; (2) A formal human evaluation study with small business owners as judges; (3) RAG integration following SCRABLE [2] to retrieve high-quality past replies as dynamic few-shot examples; (4) Evaluation of Korean-specialized LLMs such as EXAONE-3.5 as drop-in replacements for the generation stage.

REFERENCES

- [1] M. Park, J. Yun, and B. Kim, “PARAN: Persona-Augmented Review ANswering System on Food Delivery Review Dataset,” *arXiv preprint arXiv:2512.10148*, 2024.
- [2] G. Azov, T. Pelc, A. Fledel Alon, and G. Kamhi, “Self-Improving Customer Review Response Generation Based on LLMs,” in *Proc. 7th Workshop on e-Commerce and NLP (ECNLP) @ LREC-COLING*, 2024.
- [3] J. Niimi, “Dynamic Sentiment Analysis with Local Large Language Models using Majority Voting: A Study on Factors Affecting Restaurant Evaluation,” *arXiv preprint arXiv:2407.13069*, 2024.
- [4] D. Jang, S. Byun, H. Jo, and H. Shin, “KIT-19: A Comprehensive Korean Instruction Toolkit on 19 Tasks for Fine-Tuning Korean Large Language Models,” in *Proc. LREC-COLING*, 2024.
- [5] E. J. Hu *et al.*, “LoRA: Low-Rank Adaptation of Large Language Models,” in *Proc. ICLR*, 2022.
- [6] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” in *Proc. EMNLP*, 2019.
- [7] T. Zhang *et al.*, “BERTScore: Evaluating Text Generation with BERT,” in *Proc. ICLR*, 2020.
- [8] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “BLEU: A Method for Automatic Evaluation of Machine Translation,” in *Proc. ACL*, 2002, pp. 311–318.
- [9] C.-Y. Lin, “ROUGE: A Package for Automatic Evaluation of Summaries,” in *Proc. ACL Workshop on Text Summarization Branches Out*, 2004, pp. 74–81.